

Atmospheric Thermodynamics

Introduction

- Thermodynamics is defined as the study of equilibrium states of a system which has been subjected to energy transformation.

More specifically, thermodynamics is concerned with transformations of heat into mechanical work and of mechanical work into heat.

- We must first understand the way in which a gas behaves, especially when heat is added or removed. Thus, we begin by studying thermodynamics and its application in simple atmospheric contexts.
- Helps to understand the Cloud Formation and Precipitation Process & Atmospheric Dynamics.

Why study atmospheric thermodynamics?

- Our atmosphere is a mixture of gases. How this mixture behaves under spatially varying temperatures and pressures is critical to predicting weather and climate. We therefore need to know the laws that govern gases in detail.
- Solar radiation is the ultimate source of all the energy that drives our climate system. The interaction of radiation with the multiple gases that constitute our atmosphere creates a temperature profile in accord with the laws of thermodynamics.
- The spatial structure of the temperature field is related to the spatial structures of pressure and density in accord with the equation of state.
- Spatial variations of pressure drive atmospheric winds.
- The phase changes of water play an enormous role in our climate system (clouds, precipitation, ice, etc.).
- Phase changes of a substance are governed by thermodynamics.

Basic concepts of thermodynamics

Thermodynamic system

A specified collection of matter is called a system, which is defined by the mass and the composition.

- *Open system: Mass & Energy exchanges with its surroundings*
- *Closed system: Energy exchanges but NO mass with its surroundings.*
- *Isolated system: It exchanges neither mass nor energy.*

What type of system does **atmospheric thermodynamics** deal with?

The systems that atmospheric thermodynamics deal with include

1) an air parcel; 2) a cloud; 3) the ambient atmosphere; 4) an air mass etc.

Precisely speaking, they are open systems because mass can be changed by the entrainment and mixing processes.

But, we will predominantly treat them as a closed system in this course.

Gas Laws

The thermodynamic state of the atmosphere at any point is determined by the values of P, T, V, ρ at that point.

Ideal Gas Law		$PV = m R T$
---------------	--	--------------

Because $m/V = \rho$, where ρ is the density of the gas, the ideal gas equation may also be written in the form

$$P = \rho R T$$

$$P \alpha = R T$$

Where $\alpha = (1/\rho)$ is the *specific volume* of the gas, i.e., the **volume occupied by 1 kg of the gas** at pressure p and temperature T .

Gas law for Dry and Moist air

The thermodynamic state of the atmosphere at any point is determined by the values of P, T, V, ρ at that point.

Ideal Gas Law		$PV = m R T$
---------------	--	--------------

Because $m/V = \rho$, where ρ is the density of the gas, the ideal gas equation may also be written in the form

$$P_d = \rho_d R_d T \qquad P_v = \rho_v R_v T$$

Dalton's partial pressure law

$$P = (\rho_d R_d + \rho_v R_v) T$$

Where $\varepsilon = (R_d / R_v) = 0.622$ is the *mixing ratio*

➤ *Ratio of “gas constant of dry air to gas constant of moist air”*

Gas constants of Dry and Moist air

Often in meteorology, we use mass-specific gas laws so that we must specify the gas that we are talking about, Usually only dry air (N₂ + O₂ + Ar + CO₂ +...) & Water vapor (gaseous H₂O).

Gas Constant of Dry air:

$$R_d = \text{Universal Gas Constant (R}^*) / \text{Molecular weight of dry air}$$
$$= \frac{8314.5 \text{ J/K Kmol}}{28.97 \text{ Kg/Kmol}}$$

$$R_d = 287 \text{ J Kg}^{-1} \text{ K}^{-1}$$

Gas Constant of Moist air:

$$R_v = \text{Universal Gas Constant (R}^*) / \text{Molecular weight of moist air}$$
$$= \frac{8314.5 \text{ J/K Kmol}}{18 \text{ Kg/Kmol}}$$

$$R_v = 461.9 \text{ J Kg}^{-1} \text{ K}^{-1}$$

Exercises

#1:

What is the density of a dry air at the 500 hPa pressure level if the temperature is -20°C ?

#2:

If at 0°C the density of dry air alone is 1.275 kg m^{-3} and the density of water vapour alone is $4.770 \times 10^{-3} \text{ kg m}^{-3}$, what is the total pressure exerted by a mixture of the dry air and water vapour at 0°C ?

Virtual Temperature

The virtual temperature is the temperature that dry air would need to attain in order to have the same density as the moist air at the same pressure.

❖ Moist air is less dense than dry air at the same temperature and pressure

$$T_v \equiv \frac{T}{1 - \frac{e}{p}(1 - \varepsilon)}$$

$$T_v = T(1 + 0.61q)$$

Where q is specific humidity

The virtual temperature is always greater than the actual temperature.

Exercises

#3:

Consider a blob of air ($T_{\text{blob}} = 25^{\circ}\text{C}$, $q_{\text{blob}} = 10 \text{ g kg}^{-1}$) at the same pressure level as a surrounding environment ($T_{\text{env}} = 26^{\circ}\text{C}$ and $q_{\text{env}} = 1 \text{ g kg}^{-1}$). If the blob has a lower density than its environment, then does it rise?

Virtual temperature



For better analysis of temperature or an unique representation of a single equation of state on isobaric surfaces which has a mixed state of dry and moist air

The equation of state for a dry atmosphere is $p_d = \rho_d R_d T$, where R_d is the gas constant for dry air is $287 \text{ J kg}^{-1} \text{ K}^{-1}$.

The equation of state for a moist atmosphere is $e = \rho_v R_v T$, where R_v is $461 \text{ J kg}^{-1} \text{ K}^{-1}$, ρ_d is the density of dry air, and e is the vapour pressure. But, the atmosphere is rarely ever dry and is never pure water vapor. Thus, the value for R in the real atmosphere is always between 287 and 461. **Virtual temperature provides a succinct way to describe the density of a mixture of dry air and water vapor.** We can write density (as total pressure of the mixture is $p = p_d + e$) as:

$$\rho = \rho_d + \rho_v = \frac{p_d}{R_d T} + \frac{e}{R_v T} \quad (1)$$

From the definition of water vapour mixing ratio $w = \frac{\rho_v}{\rho_d} = \frac{e/R_v T}{p_d/R_d T} = \frac{R_v}{R_d} \left(\frac{e}{p - e} \right) = \varepsilon \left(\frac{e}{p - e} \right) \cong 0.622 \left(\frac{e}{p - e} \right) \quad (2)$

We can write $\frac{e}{p} = \frac{w}{w + \varepsilon} \quad (3)$

Substituting (3) in (1) $\rho = \frac{p_d}{R_d T} \left[1 - \frac{e}{p} (1 - \varepsilon) \right] \Rightarrow p_d = \rho R_d \left\{ T \left[1 - \frac{e}{p} (1 - \varepsilon) \right]^{-1} \right\} = \rho R_d \left[T \left\{ \frac{w + \varepsilon}{\varepsilon (1 + w)} \right\} \right] = \rho R_d T_v$

where $T_v = T \left(1 + \frac{w}{\varepsilon} \right) (1 + w)^{-1} \cong T \left[1 + w \left(\frac{1}{\varepsilon} - 1 \right) \right] \cong T (1 + 0.608w) \quad [\text{as } w \ll 1]$

to describe the density of a mixture of dry air, water vapor, and hydrometeors like cloud droplets, raindrops, snow and ice

$$T_v \approx T \left[1 + 0.608 w_{\text{vapor}} - (w_{\text{liq}} + w_{\text{ice}}) \right]$$

[Note: $T_v > T$]

In words, air becomes more buoyant (effectively "warmer") as w_{vapor} increases (because light H_2O replaces heavy N_2), and droplets, drops and ice concentrations decrease (because falling hydrometeors exert downward drag on the atmosphere).

*The **virtual temperature** T_v of a moist air parcel is the temperature at which a theoretical dry air parcel would have a total pressure and density equal to the moist parcel of air.*

Geopotential

- The Geopotential Φ of any point in the atmosphere is the work that must be done against the earth's gravitational field in order to raise a mass of 1 kg from sea level to the point.
- Φ is the gravitational potential per unit mass (Units: $\text{m}^2 \text{s}^{-2}$).

The force acting on 1 kg at height z above sea level is numerically equal to g . The work (in joules) in raising 1 kg from z to $z + dz$ is “ gdz ”; therefore

$$d\Phi = g \, dz$$

The geopotential $\Phi(z)$ at height z is thus given by

$$\Phi(z) = \int_0^z g \, dz$$

From the geopotential, we can also define a “Geopotential Height”

where $g_0 = 9.8 \text{ ms}^{-2}$

$$Z \equiv \frac{\Phi(z)}{g_0} = \frac{1}{g_0} \int_0^z g \, dz$$

Geopotential height is used as vertical coordinate in most atmospheric applications where energy plays an important role.

Hydrostatic Equation

Decreasing pressure with height imposes an upward force on any element of fluid (the pressure gradient force), which almost exactly balances the gravitational force due to the mass of atmosphere above.

⇒ the upward pressure gradient force acting on a thin slice of air (p decreases with z) is generally very closely in balance with the downward force due to gravitation

The **net upward** force acting on the slice of air

$$\frac{\vec{F}_{PGF}}{A} = \delta p, \quad (\delta p < 0)$$

where A is the horizontal cross section area of the slice.

The **gravitational force** acting on the slice of air

$$\frac{\vec{F}_{gravity}}{A} = \rho g \delta z$$

where g is the acceleration due to gravity (9.8 m s^{-2}),

m is the mass of this thin slice,

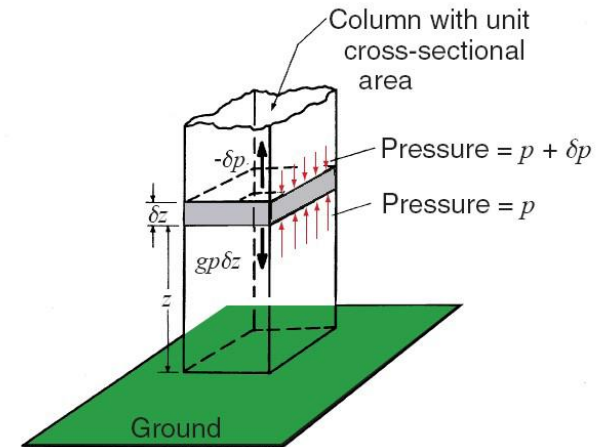
ρ is the density of the fluid.

Hydrostatic equilibrium ⇒ $\vec{F}_{PGF} = \vec{F}_{gravity}$

$$\frac{\delta p}{\delta z} = -\rho g$$

$$\lim_{\Delta p, \Delta z \rightarrow 0} \frac{\delta p}{\delta z} = \frac{dp}{dz}$$

$$\Rightarrow \frac{dp}{dz} = -\rho g$$



$$\frac{dp}{dz} \equiv -\rho g$$

This balance is referred to as the hydrostatic equilibrium

Hypsometric Equation & Scale height

Provides the thickness of an atmospheric layer between two pressure surfaces.

$$Z_2 - Z_1 = \frac{R_d \bar{T}_v}{g_0} \ln \left(\frac{p_1}{p_2} \right)$$

- It states that the thickness of a layer between two pressure surfaces is proportional to the (mean) temperature of this layer.

Where $H = \frac{RT}{g_0} = 29.3 T$ ----- Scale height

- The scale height is the height at which pressure is reduced by a factor of e (2.718)
- For the atmosphere as a whole it is about 8 km
- The atmospheric pressure falls off exponentially with height, which means that about every 8 km, the atmospheric pressure is about 1/3 less.

Exercises

#4:

Calculate the thickness of the layer between the 1000- and 500-hPa pressure surfaces

- (a) at a point in the tropics where the mean virtual temperature of the layer is 15 °C and**
- (b) (b) at a point in the polar regions where the corresponding mean virtual temperature is -40 °C.**

Types of Processes

Isobaric Processes:

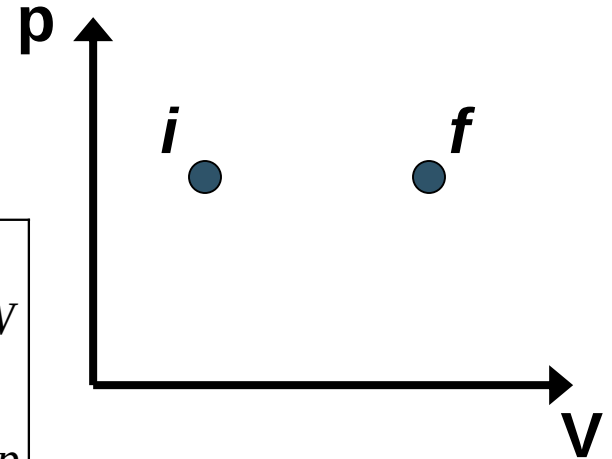
- Transformations at constant pressure
- $dp = 0$

Isobaric: $dp = 0$

Isothermal: $dT = 0$, $dQ = dW$

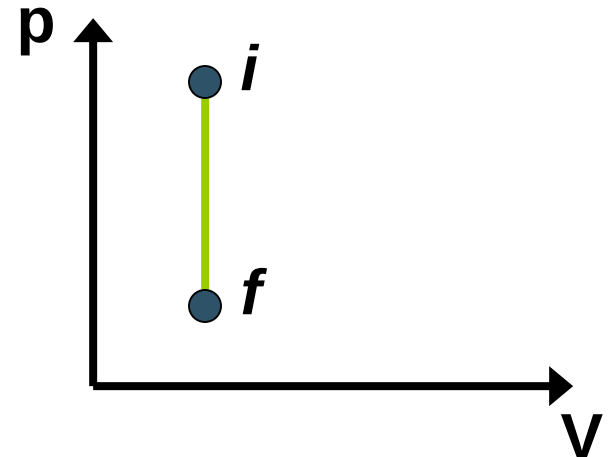
Isochoric: $d\alpha = 0$, $dQ = dU$

Adiabatic: $dQ = 0$, $c_p dT = \alpha dp$



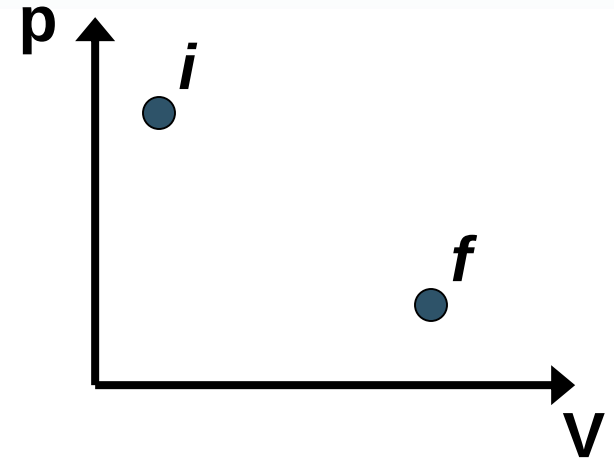
Isochoric Processes:

- Transformations at constant volume
- $dV = 0$
- $d\alpha = 0$



Isothermal Processes:

- Transformations at constant temperature
- $dT = 0$



Adiabatic Processes:

- Transformations without the exchange of heat between the environment and the system
- $dQ = 0$

Isobaric: $dp = 0$

Isothermal: $dT = 0$, $dQ = dW$

Isochoric: $d\alpha = 0$, $dQ = dU$

Adiabatic: $dQ = 0$, $c_p dT = \alpha dp$

First Law of Thermodynamics

Law of conservation of “total energy”.

Heat added per unit mass = (Change in internal energy per unit mass) +
(Work done by a unit mass)

$$\delta Q = \delta U + \delta W$$

➤ The internal energy that increases according to the heat supplied and diminishes according to the work done by the system.

The first law of thermodynamics can also be expressed as

$$dQ = dU + Pd\alpha$$

Air parcel : a small mass of air that is **thermally insulated** from its environment, such that its temperature changes **adiabatically**, and that has exactly the same pressure as its environment

Forms of the First Law of Thermodynamics

$$dq = du + dw$$

$$dq = du + pd\alpha$$

Specific Heats 

$$dq = c_v dT + pd\alpha$$

$$dq = c_p dT - \alpha dp$$

where: p = pressure

V = volume

T = temperature

α = specific volume

U = internal energy

W = work

Q = heat energy

n = number of moles

c_v = specific heat at constant volume ($717 \text{ J kg}^{-1} \text{ K}^{-1}$)

c_p = specific heat at constant pressure ($1004 \text{ J kg}^{-1} \text{ K}^{-1}$)

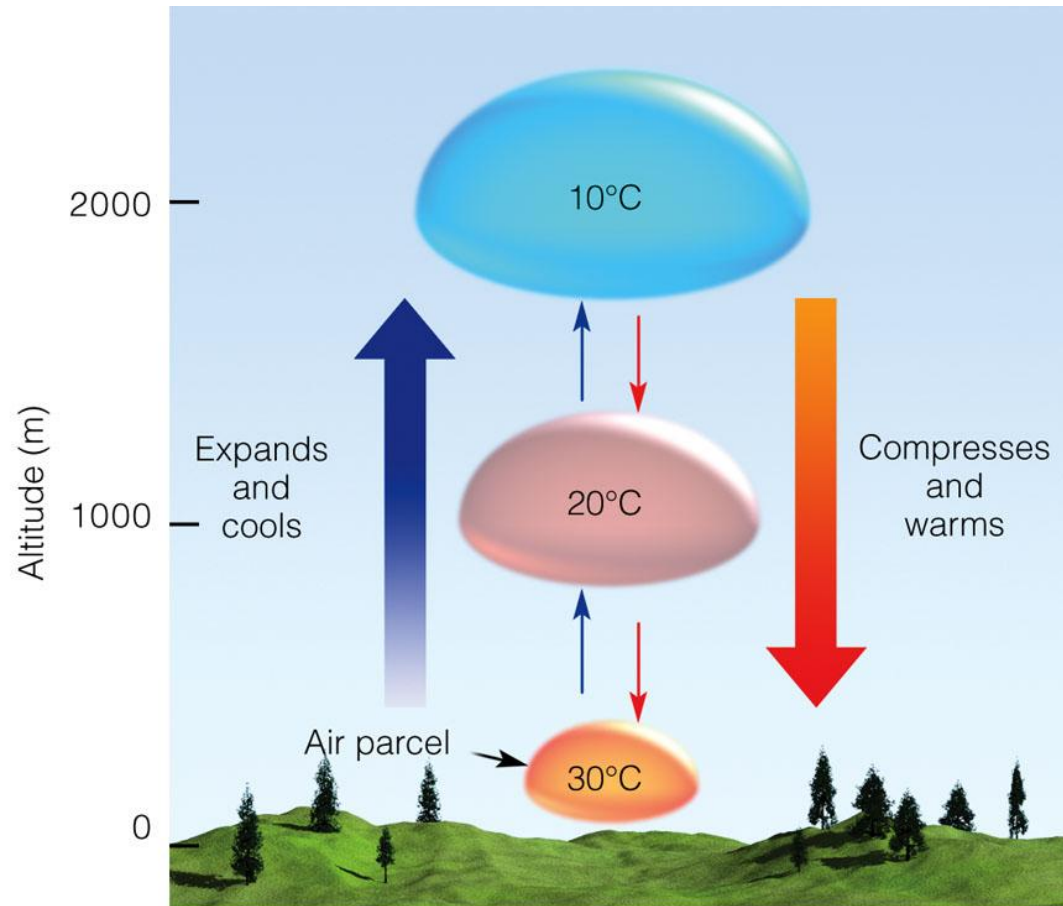
R_d = gas constant for dry air ($287 \text{ J kg}^{-1} \text{ K}^{-1}$)

R^* = universal gas constant ($8.3143 \text{ J K}^{-1} \text{ mol}^{-1}$)

Adiabatic cooling/heating

- **Ascending motion:** the parcel of air expands, does work on the rest of the atmosphere and **cools down**.
- **Descending motion:** the air parcel compresses, work is done on the air parcel and it **warms up**.
- **Adiabatic process:** **no heat** is exchanged between the air parcel and the rest of the atmosphere

This expansion/contraction causes the air to cool (warm).



© 2007 Thomson Higher Education

Air inside can freely expand or contract, but heat and air molecules do not cross the boundary.

The adiabatic process often occurs when air rises or descends and is an important process in the atmosphere.

Diabatic Process

- Involve the direct addition or removal of heat energy.
- Example: Air passing over a cool surface loses energy through conduction.

Lapse Rates

Lapse rate: The rate at which temperature changes in the vertical (slope of the temperature vertical profile)

Dry adiabatic lapse rate: the rate at which an unsaturated air parcel will cool if it rises or warm as it sinks (applies to an air parcel with a relative humidity of less than 100%) ----- **Dry adiabatic lapse rate = $9.8^{\circ}\text{C} / \text{km}$**

Moist adiabatic lapse rate: the rate at which a saturated air parcel will cool if it rises or warm if it sinks (applies to an air parcel with a relative humidity of 100%)-----**Moist adiabatic lapse rate = $6^{\circ}\text{C} / \text{km}$**

Environment: the atmosphere outside of an air parcel

Environmental lapse rate: the rate at which the environment's temperature decreases with increasing altitude

Use the environmental lapse rate to determine the temperature of the environment as you move up or down in the atmosphere.

Use the dry adiabatic lapse rate or moist adiabatic lapse rate to determine the temperature of an air parcel as it rises or sinks in the atmosphere

Dry adiabatic lapse rate

Lapse rate = rate at which temperature changes with elevation

Consider an air parcel undergoing an adiabatic change in pressure, with no phase change of any water substance in the air parcel. For this air parcel the first law of thermodynamics

can be written as:

$$c_p dT = \alpha dp = \frac{R_d T}{p} dp$$

Since this change in pressure implies a change in elevation

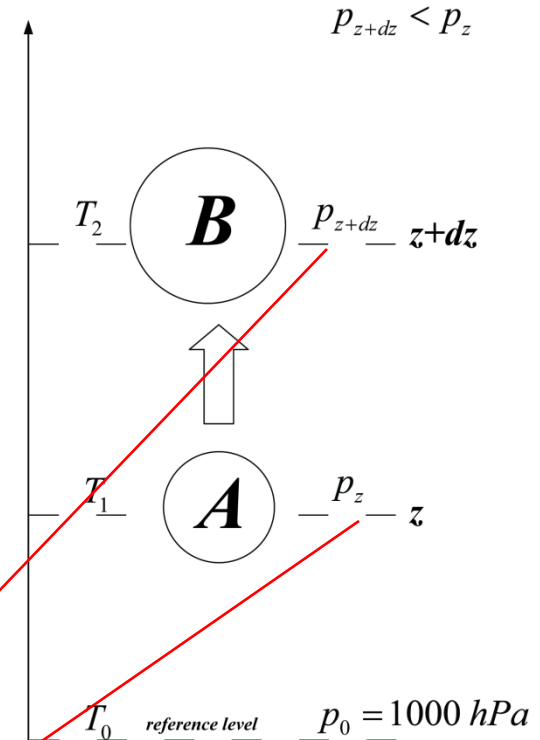
$$c_p \frac{dT}{dz} = \frac{R_d T}{p} \frac{dp}{dz} \Rightarrow \frac{dT}{dz} = -\frac{R_d T}{c_p p} (\rho g) \equiv -\frac{g}{c_p}$$

$$\left(-\frac{dT}{dz} \right)_{DA} = \frac{g}{c_p} = \frac{9.80665 \text{ m s}^{-2}}{1004 \text{ J kg}^{-1} \text{ K}^{-1}} = 9.8 \text{ K km}^{-1}$$

The lapse rate in a column of air (referred to as the environmental lapse rate) need not be equal to the dry adiabatic lapse rate and typically averages $6\text{--}7 \text{ K km}^{-1}$ in the troposphere.

In general a lapse rate always refers to the rate of decrease of temperature with height $\left(-\frac{dT}{dz} \right)$

*Adiabatic expansion
(pressure changes) – not an isobaric process*



- What is the sign of the lapse rate, if
- (i) T decreases with height
 - (ii) T increases with height
 - (iii) T is constant with height

Potential Temperature

The potential temperature θ of an air parcel is defined as the temperature that the parcel of air would have if it were expanded or compressed adiabatically from its existing pressure and temperature to a standard pressure p_0 (generally taken as 1000 hPa).

$$\theta = T \left(\frac{1000}{p} \right)^{R/c_p}$$

Equation is called Poisson's equation. It is usually assumed that $R = R_d = 287$ and $c_p = c_{pd} = 1004$; therefore, $R/c_p = 0.286$.

- Potential temperature is a conserved quantity for an air parcel that moves around in the atmosphere under adiabatic conditions.
- It is an extremely useful parameter in atmospheric thermodynamics

Potential temperature

A conserving temperature variable for adiabatic processes

Consider a parcel of air with temperature T and pressure p . Suppose that it is given a small amount of heat dQ per unit mass and as a consequence its temperature and pressure change by amounts dT and dp , respectively.

The first law of thermodynamics gives $dQ = c_p dT - \alpha dp$

where $\alpha = 1/\rho \Rightarrow$ specific volume (volume/unit mass)

c_p = specific heat at constant pressure ($= 1005.64 \text{ J kg}^{-1}\text{K}^{-1}$)

R_d = Gas constant of dry air ($= 287 \text{ J kg}^{-1}\text{K}^{-1}$)

Using perfect gas equation $p\alpha = R_d T$, eliminating α ,

$$(\text{entropy}) \quad s \equiv \frac{dQ}{T} = c_p \frac{dT}{T} - R_d \frac{dp}{p} = \frac{c_p}{R_d} (d \ln T - d \ln p)$$

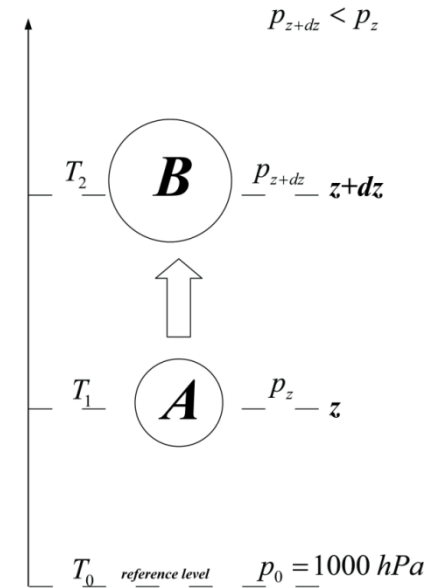
In an **adiabatic process** there is no heat input, $dQ \equiv 0$,

$$\frac{dT}{T} = \kappa \frac{dp}{p}; \quad \kappa = \frac{R_d}{c_p}$$

Integrating between reference level ($T_0 \equiv \theta$; $p_0 = 1000 \text{ hPa}$)

to any arbitrary level (T, p)

$$\left(\frac{T}{T_0} \right) = \left(\frac{p}{p_0} \right)^\kappa \Rightarrow \theta = T \left(\frac{p_0}{p} \right)^\kappa \quad \text{where } \theta = \text{potential temperature}$$



A wide range of atmospheric motions are approximately adiabatic (= isentropic motions), potential temperature is an important thermodynamic variable for such motions.

θ = measure of entropy of dry air

Exercises

#5:

If the temperature at 500 hPa is -20C, what is the potential temperature of a parcel of air at that level ?

Dry adiabatic lapse rate

- For a dry adiabatic, hydrostatic atmosphere the potential temperature θ does not vary in the vertical direction:

$$\frac{\partial \theta}{\partial z} = \frac{\partial \theta}{\partial p} = 0$$

θ is uniform in height $\Leftrightarrow$ the atmospheric column is well-mixed
 neutral conditions $\Leftrightarrow$ well-mixed
 neutral conditions $\rightarrow$ Temperature decreases by $9.8^\circ\text{C km}^{-1}$

- In a dry adiabatic, hydrostatic atmosphere the temperature T must therefore decrease with height.
- Static stability of the environment can be measured with the buoyancy frequency N .

$$N^2 \equiv g \frac{\partial \ln \theta}{\partial z}$$

$$\theta = T \left(\frac{1000}{p} \right)^{R/c_p}$$

$$\frac{\partial \theta}{\partial z} > 0 \text{ statically stable} \quad \Leftrightarrow \quad N^2 > 0$$

$$\frac{\partial \theta}{\partial z} = 0 \text{ statically neutral (dry-adiabatic)} \quad \Leftrightarrow \quad N^2 = 0$$

$$\frac{\partial \theta}{\partial z} < 0 \text{ statically unstable} \quad \Leftrightarrow \quad N^2 < 0$$

STABILITY CRITERIA

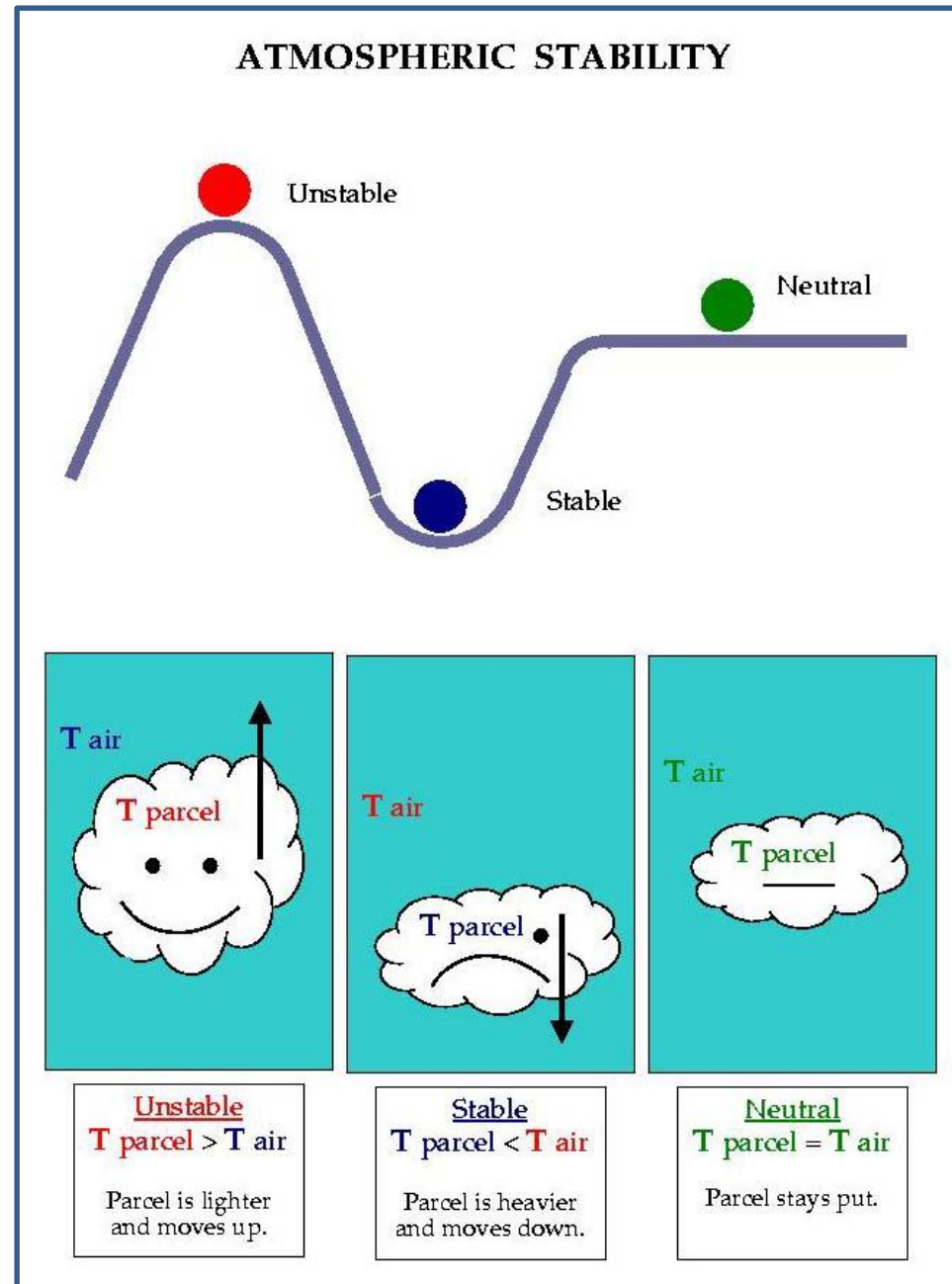
Hydrostatic stability or instability depends on the magnitude of the local environmental lapse rate (γ), relative to the dry adiabatic lapse rate (parcel lapse rate).

This result may be summarized as follows:

$\gamma_{env} > \gamma_{dry}$	$\Rightarrow$ Unstable
$\gamma_{env} = \gamma_{dry}$	$\Rightarrow$ Neutral
$\gamma_{env} < \gamma_{dry}$	$\Rightarrow$ Stable

in terms of the potential temperature lapse rate of the environment, relative to the potential temperature lapse rate of the parcel (which is zero for adiabatic changes)

$\frac{\partial \theta}{\partial z} < 0$ or $\frac{\partial \theta}{\partial p} > 0$	$\Rightarrow$ Unstable
$\frac{\partial \theta}{\partial z} = \frac{\partial \theta}{\partial p} = 0$	$\Rightarrow$ Neutral
$\frac{\partial \theta}{\partial z} > 0$ or $\frac{\partial \theta}{\partial p} < 0$	$\Rightarrow$ Stable



Moist Adiabatic Lapse Rate

- Initially, the newly saturated air will have a large rate of condensation. But as the air continues to rise, and more water vapor changes to liquid, it becomes drier. The formation of water droplets by condensation accordingly reduces as the air continues to rise. T
- The moist rate varies between about 4 and 9 C/km. It is reasonable for us to assume an "average" value of about 5 or 6 C/km.



Moist Adiabatic Lapse Rate

Consider an air parcel undergoing an adiabatic change in pressure, with no phase change of any water substance

in the air parcel. For this air parcel the first law of thermodynamics can be written as: $\frac{dT}{T} = \frac{R_d}{c_p} \frac{dp}{p}$ and

with phase changes,

one can write, $\frac{dT}{T} = \frac{R_d}{c_p} \frac{dp}{p} - \left(\frac{L}{c_p T} \right) \frac{dw_s}{T}$, where R_d = Gas constant of dry air ($= 287 \text{ J kg}^{-1} \text{ K}^{-1}$),

R_m = Gas constant of moist air ($= 461 \text{ J kg}^{-1} \text{ K}^{-1}$), $\varepsilon = \frac{R_d}{R_m} \approx 0.622$, w_s = saturation water vapor mixing ratio (kg kg^{-1}),

c_p = specific heat at constant pressure ($= 1004 \text{ J kg}^{-1} \text{ K}^{-1}$), L = Latent heat of condensation ($= 2.5 \times 10^6 \text{ J kg}^{-1}$).

Moist adiabatic lapse rate (γ_m) can be written as: (using hydrostatic and Clausius-Clapeyron equation)

$$\gamma_m = \left(-\frac{dT}{dz} \right)_m = \left[\frac{1 + \left(\frac{L w_s}{R_d T} \right)}{1 + \left(\frac{L^2 \varepsilon w_s}{R_d c_p T^2} \right)} \right] \left[\frac{g}{c_p} \right] = \left[\frac{1 + \left(\frac{L w_s}{R_d T} \right)}{1 + \left(\frac{L w_s}{R_d T} \right) \left[\frac{L \varepsilon}{c_p T} \right]} \right] \left[\frac{g}{c_p} \right]$$

One can see that $\gamma_m < \gamma_{dry}$, whenever $L\varepsilon > c_p T$

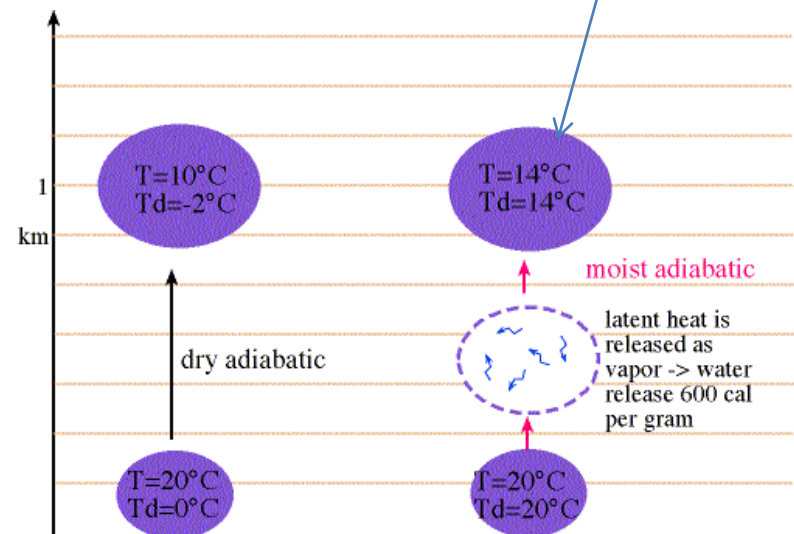
$$L = 2.5 \times 10^6 \text{ J kg}^{-1}; \quad \varepsilon = 0.622; \quad w_s = \varepsilon \left(\frac{e_s(T)}{p - e_s(T)} \right)$$

$$R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1}, \quad c_p = 1005 \text{ J kg}^{-1} \text{ K}^{-1}$$

$$\gamma_{dry} \equiv -\frac{g}{c_p} = -9.8 \text{ K km}^{-1}; \quad \gamma_m \approx -6.5 \text{ K km}^{-1}$$

$$w_s \rightarrow 0, \quad \gamma_m \rightarrow \gamma_{dry}$$

**Heat is released
within the parcel
- still adiabatic**

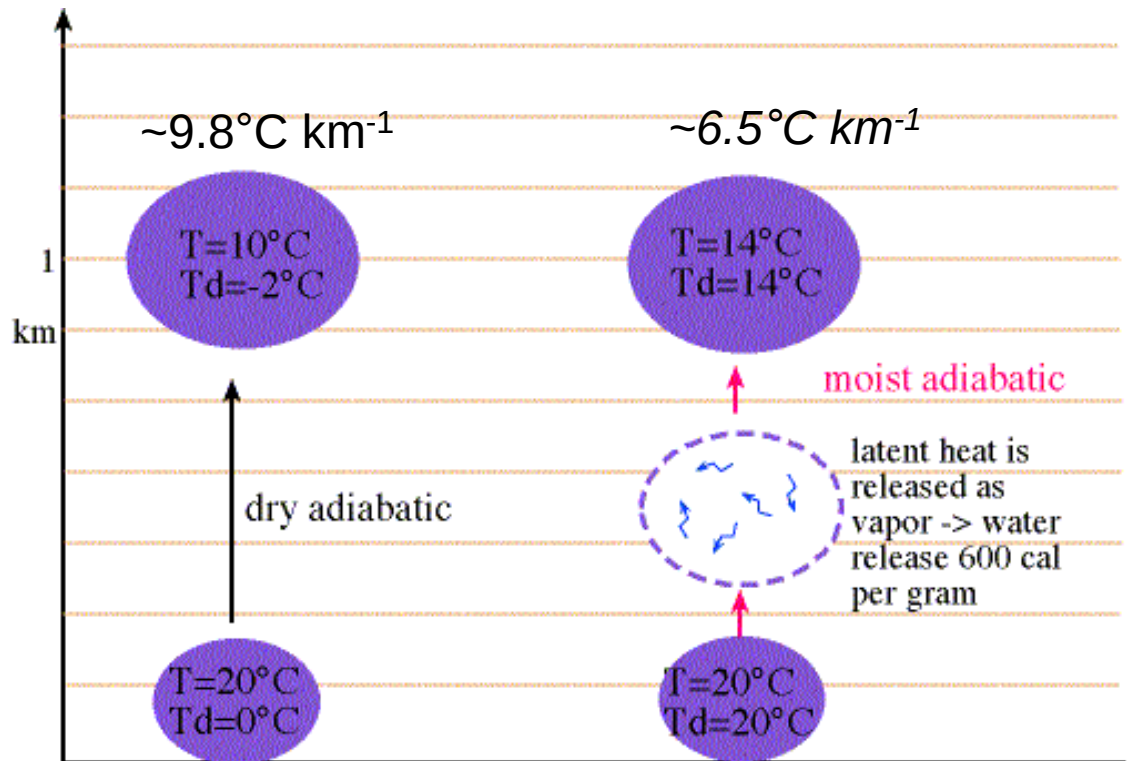


Dry vs. moist adiabatic process

Question:

Why does the parcel cool at a slower rate ($6.5^{\circ}\text{C km}^{-1}$) when it is saturated than at $9.8^{\circ}\text{C km}^{-1}$ when it is unsaturated?

The moist adiabatic lapse rate is less than the dry adiabatic lapse rate because as vapor condenses into water (or water freezes into ice) for a saturated parcel, latent heat is released into the parcel, mitigating the adiabatic cooling.

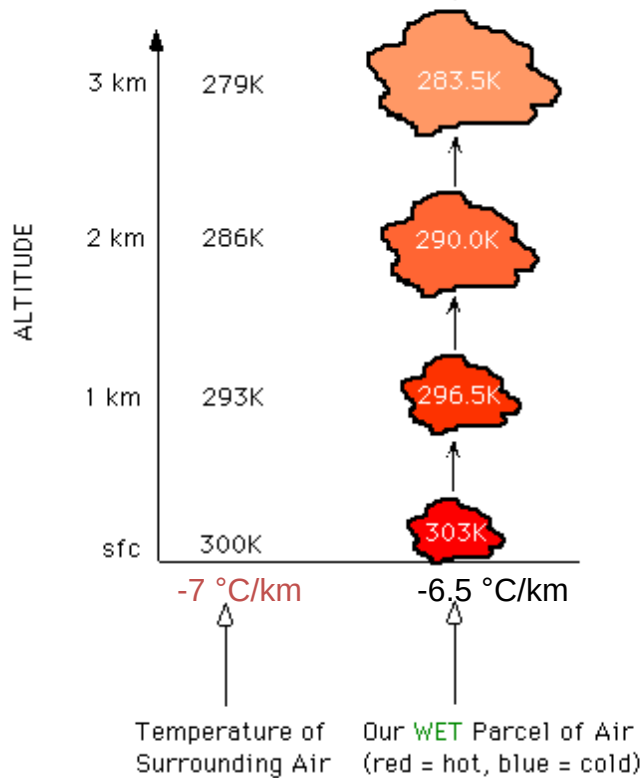


Take home message: Moisture affects stability

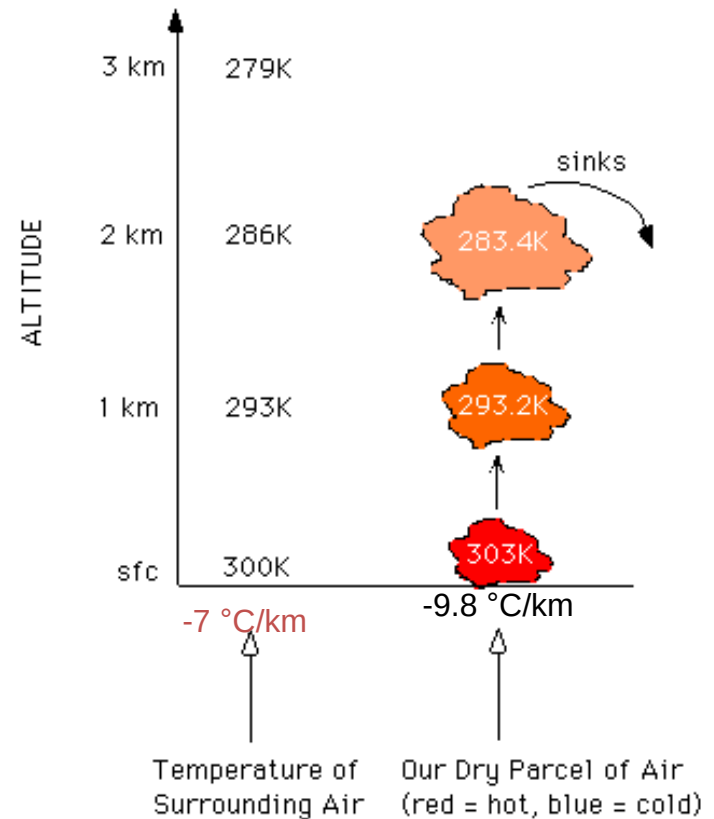
unstable



keeps rising...



stable



MOIST PARCEL rising in warm environment

DRY PARCEL rising in warm environment

Ways to quantify moisture content in the atmosphere

From the point of view of thermodynamics, the atmosphere may be considered a variable of two components, dry air and water vapour. This mixture is called moist air. There are many equivalent and widely used ways of specifying the amount of moisture in dry air.

- | | |
|--|-------------------------------|
| 1. Number density n_v [molecules m^{-3}], | 2. Partial pressure e [hPa] |
| 3. Volume mixing ratio $f_v \left[= \frac{n_v}{n_v + n_d} \right] = \frac{e}{p}$ expressed in % | |
| 4. Mass mixing ratio $w = \frac{\rho_v}{\rho_d}$ [kg kg^{-1}] $= \frac{e/R_v T}{p_d/R_d T} = \frac{R_d}{R_v} \left(\frac{e}{p - e} \right) = \varepsilon \left(\frac{e}{p - e} \right) = 0.622 \left(\frac{e}{p - e} \right)$ | |
| 5. Specific humidity $q = \frac{\rho_v}{\rho}$ [kg kg^{-1}] $\approx \frac{w}{1 + w}$ (remember: $\rho = \rho_v + \rho_d$) | |

Potential temperature (θ) of the moist unsaturated air can be defined as $\theta = T (1000/p)^{R/c_p}$, where R and c_p

correspond to moist values. The gas constants $R = \frac{k}{f_d m_d + f_v m_v} = \frac{R_d}{1 + f_v (\varepsilon - 1)} \approx R_d (1 + 0.608q)$, $k = \text{Boltzmann constant}$

Law of equipartition: the heat capacity of the mixture is the sum of the heat capacities of the components.

The specific heat capacity is just the mass-weighted mean: $c_p = (1/\rho)(\rho_d c_{pd} + \rho_v c_{pv})$ $c_p \approx c_{pd} (1 + 0.94q)$

c_{pd} (= 1005 J $\text{kg}^{-1}\text{K}^{-1}$), c_{pv} (= 1952 J $\text{kg}^{-1}\text{K}^{-1}$) are the specific heat capacities of dry air and water vapour.

$\frac{R}{c_p} = \frac{R_d}{c_{pd}} \left(\frac{1 + 0.608q}{1 + 0.94q} \right) \cong \frac{R_d}{c_{pd}} (1 - 0.33q)$	$\frac{R}{c_p}$ does not deviate from the dry value by more than 1% as q is rarely larger than about 4% in the atmosphere
---	---

Take home message: Even for moist (unsaturated) air, we can simply write $\theta = T (1000/p)^{R_d/c_{pd}}$

Moisture Parameters

1. Mixing Ratio:

The amount of water vapor in a certain volume of air may be defined as **the ratio of the mass m_v of water vapor to the mass of dry air; this is called the mixing ratio w .**

$$w \equiv \frac{m_v}{m_d}$$

The mixing ratio is usually expressed in grams of water vapor per kilogram of dry air.

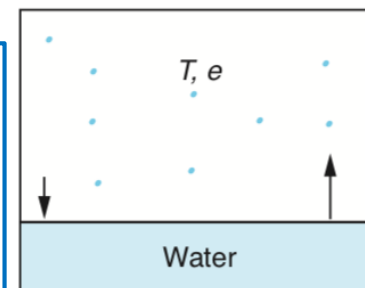
2. Specific Humidity:

The **mass of water vapor (m_v) in a unit mass of air (dry air + water vapor)** is called the specific humidity (q).

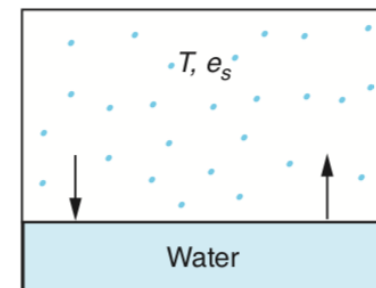
$$q \equiv \frac{m_v}{m_v + m_d} = \frac{w}{1 + w}$$

Saturation Vapour Pressure (e_s):

The rate of condensation is equal to the rate of evaporation, then the air said to be saturated and the pressure e_s that is then exerted by the water vapor is called the **saturation vapor pressure**.



(a) Unsaturated



(b) Saturated

3. Saturation Mixing Ratio:

It is defined as the ratio of **the mass m_{vs} of water vapor** in a given volume of air that is saturated to **the mass m_d of the dry air**.

$$w_s \equiv \frac{m_{vs}}{m_d}$$

The mixing ratio is usually expressed in grams of water vapor per kilogram of dry air.

4. Relative Humidity (RH):

The ratio of the actual mixing ratio (w) of the air to the saturation mixing ratio (w_s) at the same temperature and pressure.

$$RH \equiv 100 \frac{w}{w_s} \simeq 100 \frac{e}{e_s}$$

5. Dew point temperature (T_d):

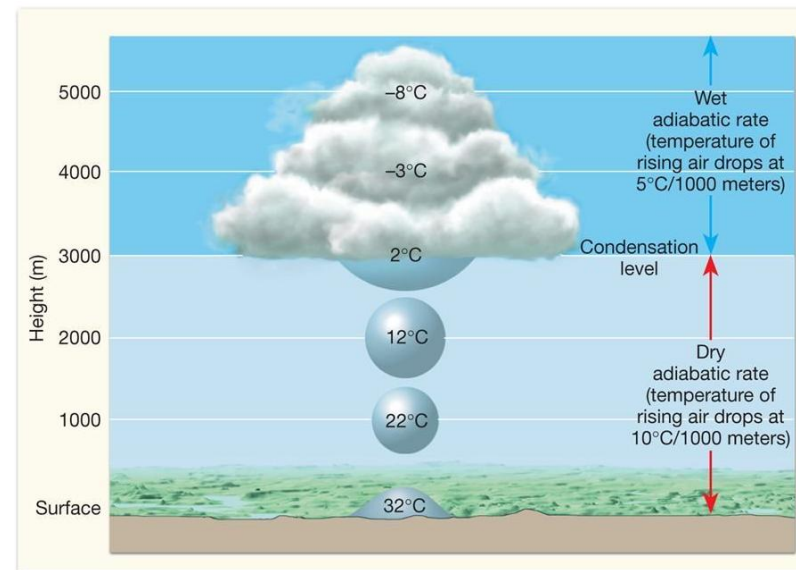
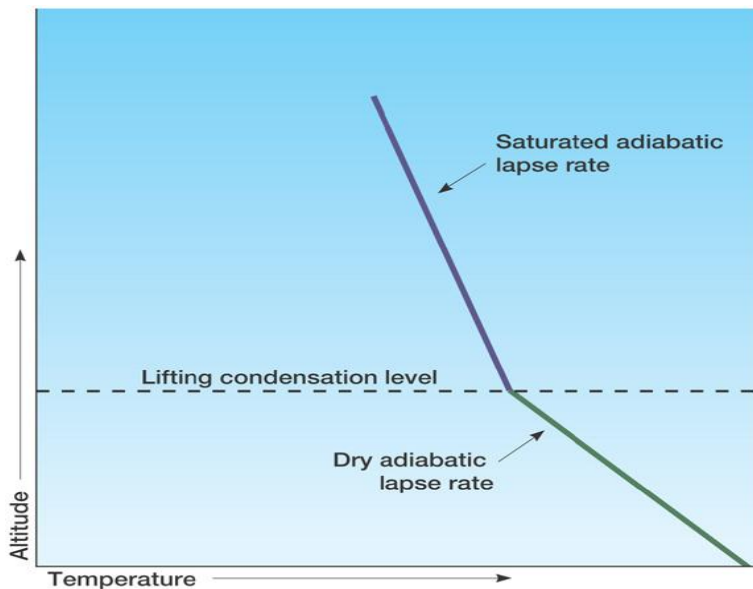
T_d the temperature to which a parcel must be cooled (e.g. by radiation) in order to be saturated. In other words, the is T_d **the temperature at which the saturation mixing ratio w_s becomes equal to the actual mixing ratio w** .

6. Wet bulb temperature:

The wet bulb temperature T_w is the temperature to which air may be cooled at constant pressure by evaporation of water into it until saturation is reached.

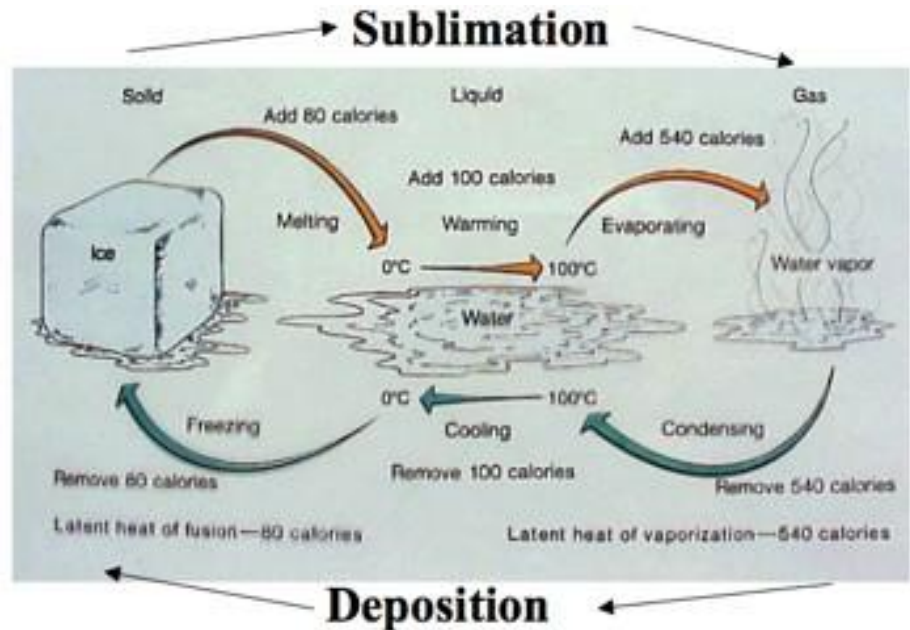
7. Lifting condensation level (LCL)

The lifting condensation level (LCL) is defined as the level to which an unsaturated (but moist) parcel of air can be lifted adiabatically before it becomes saturated.



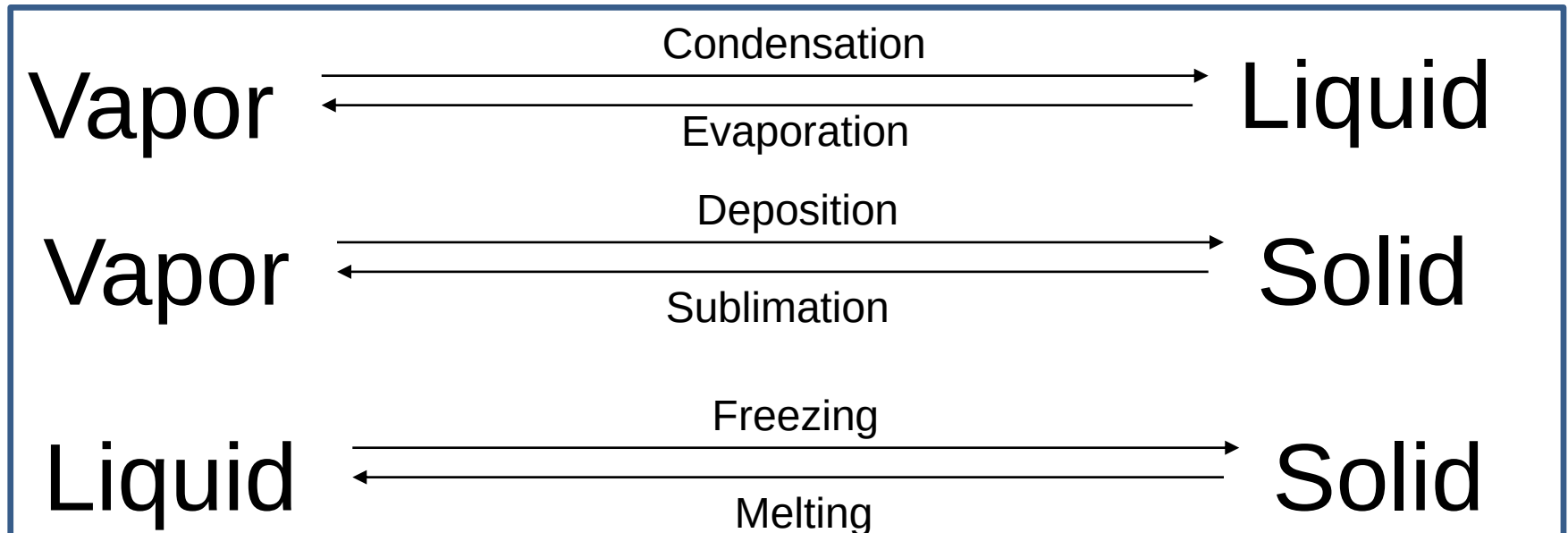
Water Vapour in Air - **Phases of water**

- **Condensation**
- **Evaporation**
- **Melting**
- **Freezing**
- **Sublimation (solid-vapor)**
 - Molecules have enough energy to escape from the surface of ice into air above and directly into the vapour phase
- **Deposition (vapor-solid)**
 - Water vapour molecule attaches itself to an ice crystal and changes to ice



Water Vapour in Air - **Phases of water**

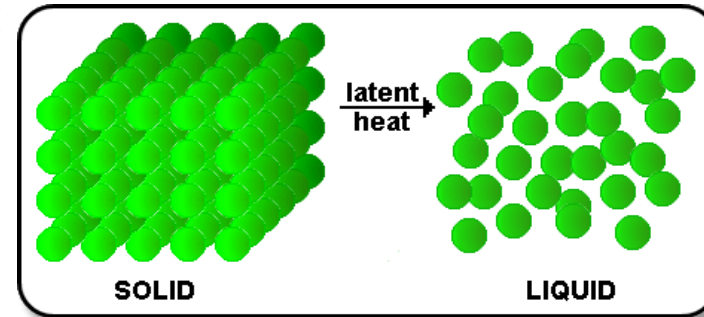
Direction of phase change	Thermodynamic effect
Going to lower energy phase (vapor → liquid → ice) Examples: rain, ice-formation	heat is released (warms air) (condensation: exothermic)
Going to higher energy phase (ice → liquid → vapor) Examples: ice-melting, evaporation	heat is absorbed (cools air) (evaporation: endothermic)



Latent heat:

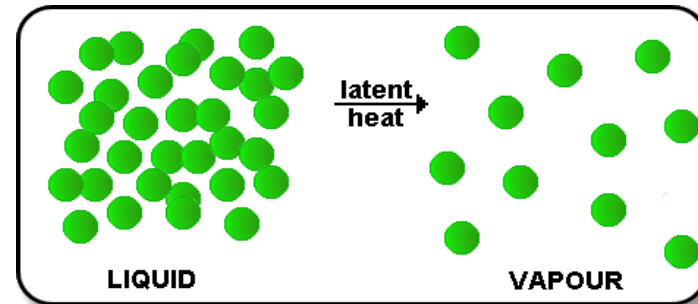
Latent heat is defined as the **heat absorbed or released** when **a substance changes its physical state** completely at constant temperature

- **Latent heat Melting** is defined as the heat that has to be given to a unit mass of a material to convert it **from the solid to the liquid phase** without a change in temperature. The particles are free from their fixed positions and are able to vibrate and move among each other .



Latent heat of melting/freezing is $3.34 \times 10^5 \text{ J kg}^{-1}$

- **Latent heat of vaporization** is the **heat absorbed** when a **liquid changes into vapor** at **constant temperature**. When liquid boils, the heat absorbed is used to completely break the bonds of the particles and also to do work against atmospheric pressure when the gaseous vapor expands into the atmosphere.



Latent heat of vaporization is $2.25 \times 10^6 \text{ J kg}^{-1}$

Clausius-Clapeyron equation

The Clausius-Clapeyron equation relates the latent heat (heat of transformation) of vaporization or condensation to the **rate of change of vapour pressure with temperature during a phase change.**

Provides the mathematical relationship that describes any equilibrium state of water as a function of temperature and pressure. Accounts for phase changes at each equilibrium state (each temperature).

$$\frac{de_s}{dT} = \frac{L_v}{T(\alpha_2 - \alpha_1)}$$

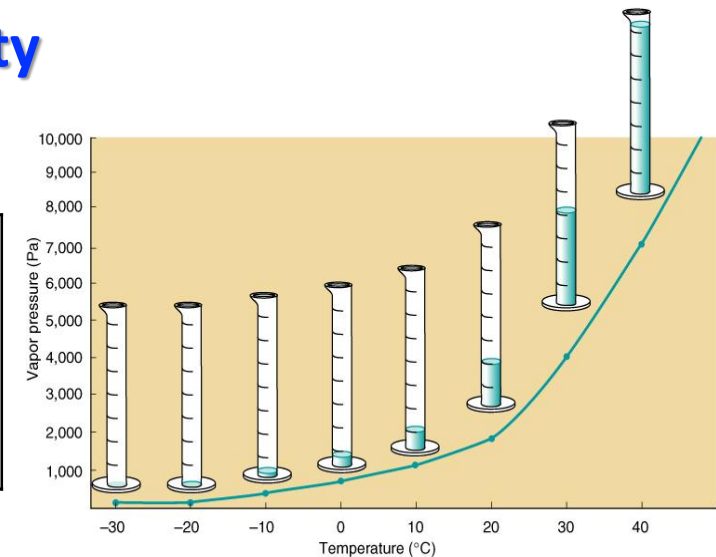
Saturation Water Vapor pressure vs. Temperature

Maximum water vapour capacity increasing as temperature increases.

Take home message:

Warmer air holds more moisture

Colder air loses the capacity of holding more moisture



Copyright 1999 John Wiley and Sons, Inc. All rights reserved.

$$e_{sw} \text{ (hPa)} = 6.11 \exp \left[\frac{L_v}{R_v} \left(\frac{1}{273.15} - \frac{1}{T(\text{K})} \right) \right]$$

FORMS OF ATMOSPHERIC ENERGY

Transformations from thermal and potential energy to kinetic energy are responsible for setting the atmosphere into motion. They maintain the circulation against frictional dissipation.

Thermal energy, which is represented in the temperature and humidity of air.

Gravitational potential energy, which is represented in the horizontal distribution of atmospheric mass.

The thermal energy of moist air has two contributions :

1. Sensible heat content reflects the molecular energy of dry air, which is the dominant component of moist air.
2. Latent heat content reflects the energy that can be imparted to the dry air component when the vapour component condenses.

These contributions are collected in the specific enthalpy

$$h = (c_p T + gz) + L_v w = \text{Sensible heat content} + \text{gravitational PE} + \text{Latent heat content}$$

$$\text{We can also write, } \frac{d}{dt} \left([c_p T + gz] + L_v w \right) = \dot{q}_{net} = \dot{q}_{rad} + \dot{q}_{mech}$$

$c_p T + gz + L_v w \Rightarrow$ moist static energy of the air parcel, which includes thermal as well as gravitational potential energy.

A parcel's moist static energy changes only through heat transfer with its environment. (involves only internal exchanges sensible and latent heat contents $\dot{q}_{mech} = \dot{q}_{sen} + \dot{q}_{latent}$. Mechanical heat transfer $\Rightarrow$ turbulent heat transfer/mixing.

A parcel acquires moist static energy at low latitudes through absorption of radiative, sensible and latent heats from the earth's surface. Upon reaching middle and high latitudes, the parcel will lose moist static energy through radiative cooling to space.

$$\text{Under thermal equilibrium } \overline{\dot{q}_{rad}} + \overline{\dot{q}_{sen}} + \overline{\dot{q}_{latent}} = 0$$

Inside the boundary layer, turbulent diffusion of heat and absorption of LW radiation from the Earth's surface increase an air parcel's sensible heat content. Absorption of water vapor from the surface of ocean increases its latent heat content. Both increase the moist static energy of the parcel. Outside the boundary layer and convection, turbulent diffusion is small enough to be ignored. Radiative cooling then becomes the primary diabatic process. It gradually depletes the parcel's store of moist static energy, which is subsequently replenished when it returns to regions of positive heat transfer.

For more details, read : Salby, Murray (2012) : Physics of the atmosphere and climate, Cambridge University Press, 665pp.

Dry static energy

Adiabatic process is defined as one for which $dq = 0$

$$\begin{aligned} dq &= c_v dT + p dv \\ dq &= c_p dT - \alpha dp \end{aligned}$$

In an adiabatic, (isochoric) process $\rightarrow$ internal energy ($u = c_v T$) is conserved.

In an adiabatic, (isobaric) process $\rightarrow$ enthalpy ($h = c_p T$) is conserved.

Many processes in the atmosphere are nearly isobaric, the physical quantity of relevance in our applications is often enthalpy rather than internal energy.

The dry-adiabatic lapse rate is: $\gamma_{dry} = -\left(\frac{dT}{dz}\right) = \frac{g}{c_p}$

$$h = c_p T \Rightarrow \frac{dh}{dz} = -g$$

The enthalpy of the parcel decreases as it rises because of the gravitational work it does.

We can also write $dh = -g dz \Rightarrow d(h + gz) = 0$ or $\frac{d}{dz}(c_p T + gz) = 0$

The quantity $c_p T + gz$ is sometimes called the "dry static energy"

Dry static energy is conserved in the process of adiabatic ascent (or descent) of a parcel.

For well mixed layers of the atmosphere, far from surfaces, in which phase changes are not taking place, the dry adiabatic lapse rate often is the lapse rate of air temperature.

Moist static energy

If the ascent (or descent) is isentropic (reversible, and adiabatic), then $\frac{dT}{dz} = -\frac{g}{c_p} - \frac{L_v}{c_p} \frac{dw_s}{dz}$

For a rising parcel in which condensation occurs, the saturation mixing ratio w_s decreases with increasing z . As a consequence, the lapse rate of such parcel is less than the dry adiabatic lapse rate. We can also write, $c_p dT = -gdz - d(L_v w_s)$

$$d(c_p T + gz + L_v w_s) = 0 \Rightarrow \frac{d}{dz}(c_p T + gz + L_v w_s) = 0$$
$$h_m = \text{moist static energy (MSE)} = c_p T + gz + L_v w_s$$

For adiabatic displacements of a moist air parcel, with no internal ice production or melting MSE is conserved. MSE is an energy, and θ_e is an entropy proxy. MSE is functionally equivalent to $c_p \theta_e$. Similarly, dry static energy is functionally equivalent to $c_p \theta$.

For most purposes, one just should use the more convenient choice.

MSE if working in height coordinates or θ_e if working in pressure coordinates.

In skew T-ln p or tephigrams, **dry adiabats based on constant dry static energy, and the moist adiabats based on saturated θ_e and MSE**

Static stability

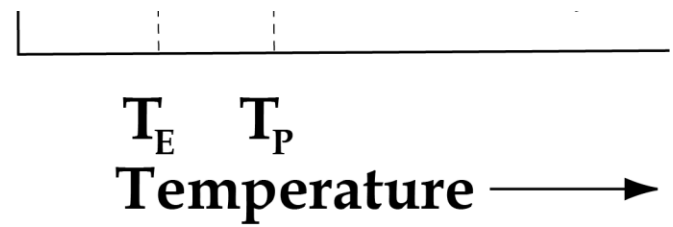
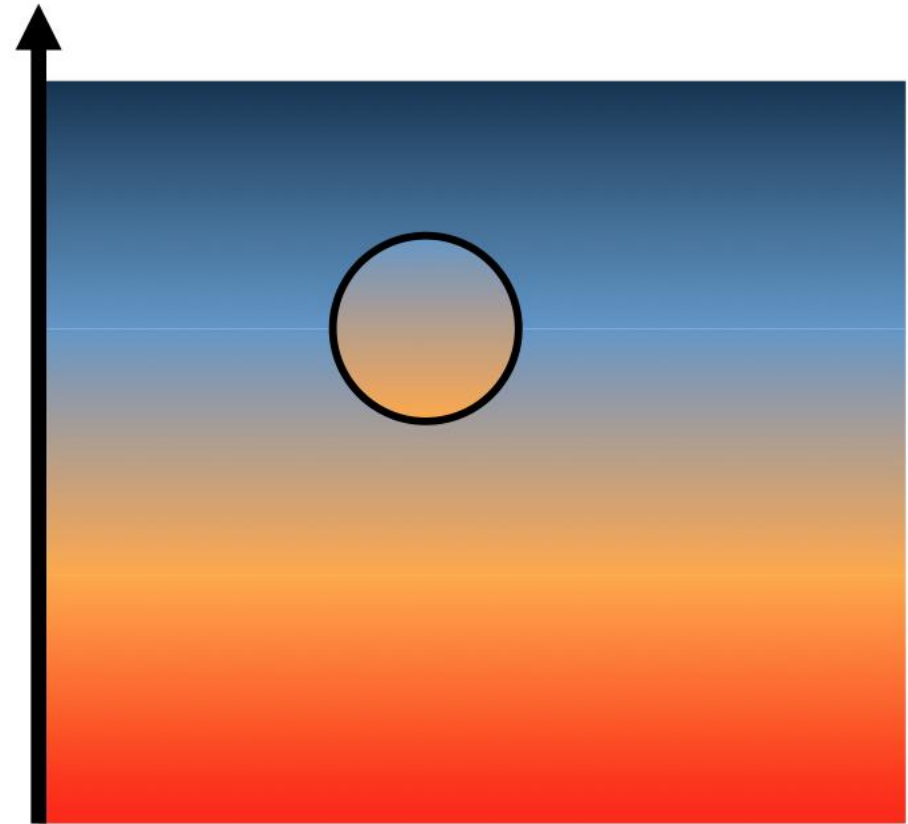
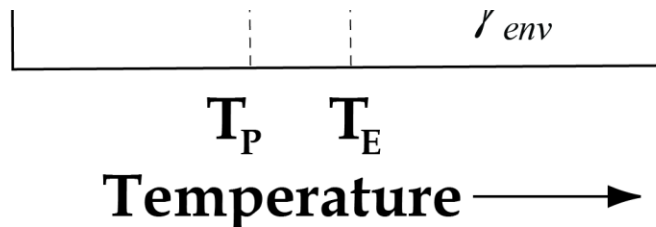
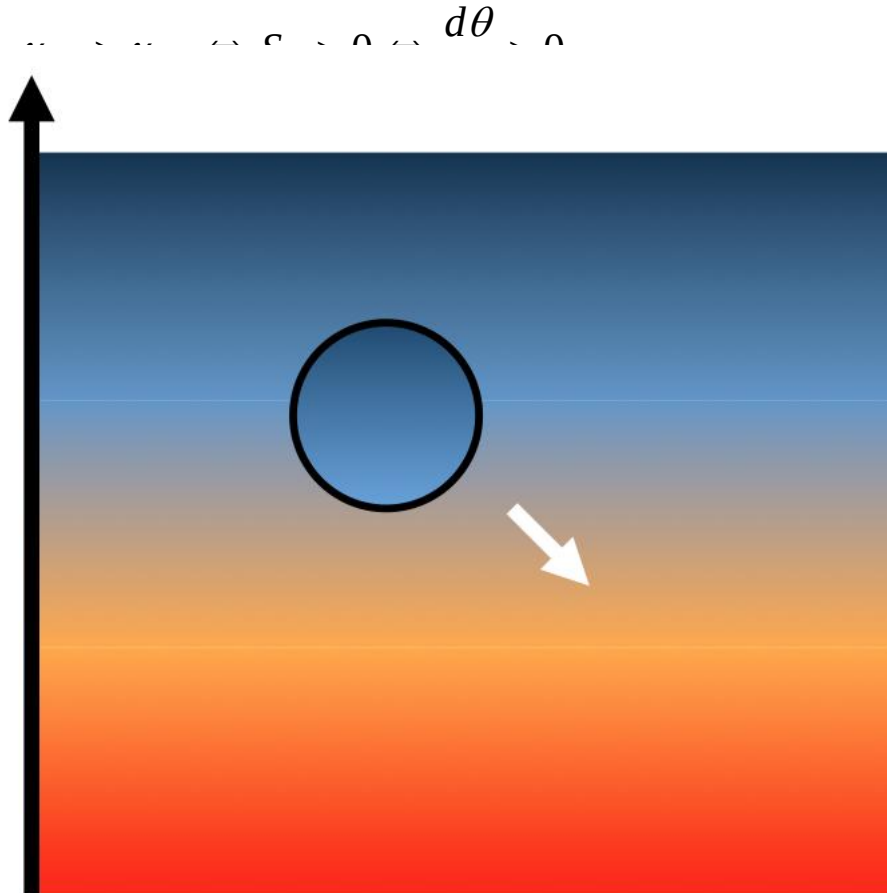
$$\frac{d\theta}{dz} = \frac{\theta}{T}(\gamma_{dry} - \gamma_{env})$$

$$S_p \equiv \frac{\gamma_{dry} - \gamma_{env}}{\rho g}$$

$$\frac{d\theta}{dz} = \frac{\theta}{T}(\gamma_{dry} - \gamma_{env})$$

$$S_p \equiv \frac{\gamma_{dry} - \gamma_{env}}{\rho g}$$

$$\gamma_{dry} < \gamma_{env} \Leftrightarrow S_p < 0 \Leftrightarrow \frac{d\theta}{dz} < 0$$



Equivalent potential temperature

The equivalent potential temperature (θ_e) is the potential temperature that would result if all of the water in the air parcel were condensed and rained out by raising the air upwards and then lowering the air parcel to the surface dry adiabatically.

The θ_e of an air parcel would be conserved if the condensed water did not rain out but instead stayed with the air parcel. For reversible moist adiabatic processes, that is processes where the condensed water does not fall out of the air parcel $\rightarrow \theta_e$ is conserved.

$$\frac{dQ}{T} = c_p \frac{d\theta}{\theta} = c_p \frac{dT}{T} - R \frac{dp}{p}$$

Latent heat associated with condensation into raising the air parcel temperature $dQ_s = -L_v dw_s$

$$c_p \frac{d\theta}{\theta} = -\frac{L_v dw_s}{T} \Rightarrow \text{Given } d \ln \theta \approx -d \left(\frac{L_v w_s}{c_p T} \right) \quad -\frac{L_v(T_2)w_s(T_2)}{c_p(T_2)T_2} + \frac{L_v(T_1)w_s(T_1)}{c_p(T_1)T_1} = \ln \theta(T_2) - \ln \theta(T_1)$$

At the extremely low w_s which occurs in very cold air in the high altitudes where there is virtually no water

left to condense out, $\theta_e = \theta$. We use this limit to do the following $\frac{L_v(T_1)w_s(T_1)}{c_p(T_1)T_1} = \ln \theta_e(T_2) - \ln \theta(T_1)$

where T_2 represents the temperature of the air parcel after it has been lifted into the upper troposphere which is sufficiently cold to set the $w_s(T_2) \rightarrow 0$. Therefore, approximate relation is given by

$$\text{Given that } \frac{dw_s}{w_s} \gg \frac{dT}{T}, \quad d \left(\frac{w_s}{T} \right) \approx \frac{dw_s}{T}, \quad d \ln \theta \approx -d \left(\frac{L_v w_s}{c_p T} \right)$$

$$\theta_e = \theta \exp \left(\frac{L_v w_s}{c_p T} \right)$$

Equivalent potential temperature (θ_e)

Let's continue with our example of the air lifting up the side of the mountain. If the mountain is high enough, and the parcel or layer of air continues to lift, all of the moisture in the lifting air could condense and fall out of the air in the form of precipitation.

Once this occurs, θ of the now dry air is considered as θ_e of the original moist parcel before condensation began.

Like the θ and the LCL, the θ_e will also be a key player when we discuss vertical stability. The θ_e will help us determine the stability of a thick layer of air versus the stability of a single isolated parcel.

$$\begin{aligned}\frac{\partial \theta_e}{\partial z} < 0 &\Rightarrow \text{convectively unstable} \\ \frac{\partial \theta_e}{\partial z} = 0 &\Rightarrow \text{neutral convection} \\ \frac{\partial \theta_e}{\partial z} > 0 &\Rightarrow \text{convectively stable}\end{aligned}$$

